

Successful RDP Brute-Force Detection Incident Report

Detection validation using Kali Linux, Windows 11 and Wazuh

Project	HOME SOC LAB
Scenario	Controlled Successful RDP brute-force simulation
Status	Detection successfully validated
Test date	22 August 2026

1. Executive Summary

A controlled brute-force authentication attack was performed against the Windows 11 SOC machine from the Kali Linux machine. The attack generated multiple failed authentication attempts, which were successfully detected by Wazuh. Following the failed attempts, a successful Remote Desktop Protocol (RDP) authentication was established.

The successful authentication was confirmed through:

- Windows Security Event ID: 4624
- Logon Type: 10
- Wazuh Rule: 92657
- Detection: Successful Remote Logon Detected

This demonstrates that the HOME SOC Lab can detect both the **failed authentication phase** and the subsequent **successful remote authentication phase**.

2. Lab Environment

Component	Role	Host-Only IP
Kali Linux	Attack simulation / RDP test source	192.168.56.105
Windows 11	Target / Windows Security telemetry source	192.168.56.106
Wazuh	SIEM / detection and alerting	Wazuh server in lab

Network safety: the lab uses a Host-Only adapter for the attack path between the VMs. The test was bounded to the Windows VM and was not directed at the user's host machine.

3. Objective

- Validate that remote RDP authentication failures generate Windows Event ID 4625.
- Confirm that the source IP is preserved as the Kali VM address rather than 127.0.0.1.
- Validate Wazuh's native failed-login and multiple-login-failure detections.
- Validate Wazuh's Successful login from an unrecognized IP
- Collect evidence for the HOME SOC LAB incident report before developing a custom correlation rule.

4. Attack Simulation

RDP was enabled on the Windows 11 VM and TCP/3389 was verified as reachable from Kali over the Host-Only network. A single manual failed login was first used to validate telemetry. After confirming that Windows recorded Event ID 4625 with the Kali source address, a bounded Hydra test was performed using six deliberately incorrect passwords.

Target: 192.168.56.106:3389 (RDP)

Source: 192.168.56.105 (Kali Linux)

Test volume: 6 intentionally failed authentication attempts

Tool: Hydra, constrained to the small lab test list

Next, after that phase when we got hands of the Password for the User by bruteforcing, we performed an RDP session, which was Successful and was detected by Wazuh, with EventID = 4624 and logon type = 10

5. Detection & Telemetry

Telemetry	Rule / Event	Severity	Observed Result
Windows Security	4625	Authentication failure	Observed; source network address mapped to Kali
Windows Security	4624	Successful Login	Observed : Source network address mapped to Kali
Wazuh	60122	Level 5	6 alerts: Logon Failure - Unknown user or bad password
Wazuh	60204	Level 10	1 alert: Multiple Windows Logon Failures

6. Incident Timeline

1. RDP connectivity verified from Kali to the Windows VM.
2. One controlled failed RDP authentication confirmed Windows Event ID 4625 and the Kali source address.
3. Six bounded failed authentication attempts were generated with Hydra.

- 4. Windows Security recorded the failed authentication activity.
- 5. Wazuh generated six Rule 60122 alerts.
- 6. Wazuh generated Rule 60204 (Level 10) for multiple Windows logon failures.
- 7. After multiple Failed Login , there was a Successful RDP Login detected by Wazuh with EventID 4624 and Logon Type 10 .
- 8. The IP belonged to the controlled Kali Source Address
- 9. Evidence was captured for the project report and detection-engineering baseline.

7. Evidence / Findings

The Wazuh Events view showed the following detection sequence:

- Six Rule 60122 events: "Logon Failure - Unknown user or bad password".
- One Rule 60204 event: "Multiple Windows Logon Failures" at Level 10.
- One Rule 92653 event : “User: WORKGROUP\vbouser logged using Remote Desktop Connection (RDP) from ip:192.168.56.105” at level 3
- The earlier single-login validation confirmed Source Network Address = 192.168.56.105.

Screenshot placeholder — insert Wazuh Events screenshot showing Rules 60122 and 60204.

Screenshot placeholder — insert Windows Event Viewer screenshot showing Event ID 4625 and the Kali source address.

8. Detection Analysis

The test demonstrates a complete detection chain. Individual authentication failures are represented by Rule 60122, while Wazuh's Rule 60204 provides a higher-severity indication that multiple Windows logon failures occurred. And the very next Rule Id 92657 detected a Remote logon , and Rule ID 92653 detected a Successful RDP .

9. MITRE ATT&CK Mapping

Technique	Name	Relevance
T1110	Brute Force	Repeated authentication failures were simulated to validate brute-force detection.
T1021.001	Remote Services: RDP	RDP was the authentication service used for the controlled lab test.

10. Impact / Risk

In a production environment, repeated failed RDP authentication attempts can indicate password guessing, credential stuffing, or an attempt to gain unauthorized remote access. And the very next moment a Successful login from the same IP indicate a possible compromise of the System . The controlled lab activity caused no intended impact to production systems; it was performed solely to validate security monitoring.

11. Response / Containment

- The RDP authentication activity was conducted within the isolated **HOME SOC LAB** environment.
- The Windows 11 SOC VM was monitored through Wazuh throughout the exercise.
- The activity was intentionally generated for detection and investigation validation and was not directed at the host machine.
- The successful RDP authentication was identified and documented using Windows Event ID **4624 / Logon Type 10** and Wazuh Rule **92657**.
- In a real-world incident, the SOC analyst should validate the source IP, affected account, and subsequent activity following the successful login.
- If the successful authentication is determined to be unauthorized, appropriate containment actions would include **protecting or disabling the affected account, blocking the malicious source, terminating the active RDP session, and isolating the affected endpoint if required**.
- Further investigation should review **post-login process creation, Sysmon telemetry, persistence mechanisms, and other endpoint activity** to determine whether additional compromise occurred.

12. Evidences

timestamp	agent.name	rule.description	rule.level	data.win.ev...	data.win.ev...	data.win.ev...	data.win.eve...	rule.id
Aug 22, 2026 @ 14:47:10.1...	Win-11-SOC	Multiple Windows Logon Failures	10	-	-	-	-	60204
Aug 22, 2026 @ 14:47:08.3...	Win-11-SOC	Logon Failure - Unknown user or bad pass...	5	-	-	-	-	60122
Aug 22, 2026 @ 14:47:06.0...	Win-11-SOC	Logon Failure - Unknown user or bad pass...	5	-	-	-	-	60122
Aug 22, 2026 @ 14:47:03.9...	Win-11-SOC	Logon Failure - Unknown user or bad pass...	5	-	-	-	-	60122
Aug 22, 2026 @ 14:47:01.9...	Win-11-SOC	Logon Failure - Unknown user or bad pass...	5	-	-	-	-	60122
Aug 22, 2026 @ 14:46:59.9...	Win-11-SOC	Logon Failure - Unknown user or bad pass...	5	-	-	-	-	60122
Aug 22, 2026 @ 14:46:57.2...	Win-11-SOC	Logon Failure - Unknown user or bad pass...	5	-	-	-	-	60122

Successful Login Detected

Aug 22, 2026 @ 14:48:14.6...	Win-11-SOC	User: WORKGROUP\vbouser logged using...	3	-	-	-	-	92653
Aug 22, 2026 @ 14:48:14.3...	Win-11-SOC	User: WORKGROUP\vbouser logged using...	3	-	-	-	-	92653
Aug 22, 2026 @ 14:47:23.4...	Win-11-SOC	Successful Remote Logon Detected - User:...	6	-	-	-	-	92657
Aug 22, 2026 @ 14:47:12.2...	Win-11-SOC	Successful Remote Logon Detected - User:...	6	-	-	-	-	92657

data.win.system.channel	Security
data.win.system.computer	Win11
data.win.system.eventID	4624
data.win.system.eventRecordID	98579
data.win.system.keywords	0x8020000000000000
data.win.system.level	0
data.win.system.message	Logon Information: Logon Type: 10 Restricted Admin Mode: No Remote Credential Guard: No Virtual Account: No Elevated Token: No

13. Conclusion

The HOME SOC Lab successfully demonstrated a controlled authentication attack scenario in which repeated failed authentication attempts were followed by a successful RDP authentication.

Wazuh successfully detected the failed authentication activity and the subsequent successful remote logon, while Windows Event ID 4624 with Logon Type 10 provided endpoint-level confirmation of the RDP session.

Stage completed:

Brute Force → Multiple Failed Logins → Successful RDP Authentication